

Modern Website Design with React



Since people are becoming increasingly aware of the significance of yoga following the pandemic, why don't we create a website and educate them more?

Yes, great idea!



Introduction

In this lesson, we are going to get familiar with the React library by creating a simple website on Yoga. This lesson will introduce you to React and the component-based design approach.

First, we'll look at how to create user interfaces using React components, as well as how to set up and run a React application. We will also divide the single page website into multiple sections, where each section will represent a React component. This lesson is designed to help you get started with React and component-based design approach.

What is React?

React is one of the widely used JavaScript libraries or frameworks, used in frontend development. Developed by Facebook, it is used to create user interfaces, also referred to as the 'View Layer of Web Applications'.

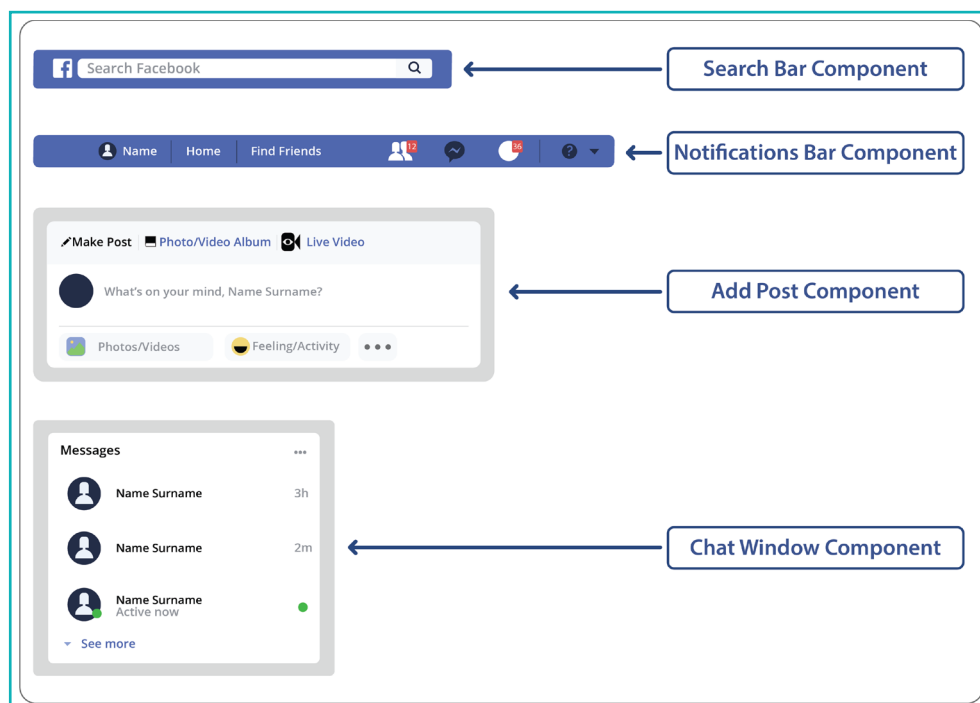
Please note that **React** is not a programming language, but rather an open-source, component-based library that is only responsible for the view layer of the application, whether it is a mobile app or a web app.

React is used to create user interfaces for popular web applications, such as Instagram and Facebook.



React Components

Observe the following image -



- You may be familiar with this image, which depicts the Facebook interface but is broken down into individual components.
- **Component-based design** refers to a design format in which the user interface is divided into multiple components, each of which has its own design and functionality.
- ‘Components’ are considered the heart of every React application.
- They form the core building blocks of React applications and facilitate the development of user interfaces.
- It is also reusable. In other words, it is not linked to a particular webpage and can be used on different pages of your web app. For example, you can create a ‘Navigation Bar’ component and use it across multiple pages on your website or web application.
- Component-based design is also effective for error detection since we debug only the specific component that has an error, rather than having to troubleshoot the entire web application.

For example, if the navigation items are not aligned correctly, then we only need to check that specific component.

Also, if the posts are not displayed, then we can just check the ‘Post’ component, instead of going through the entire application to identify errors.

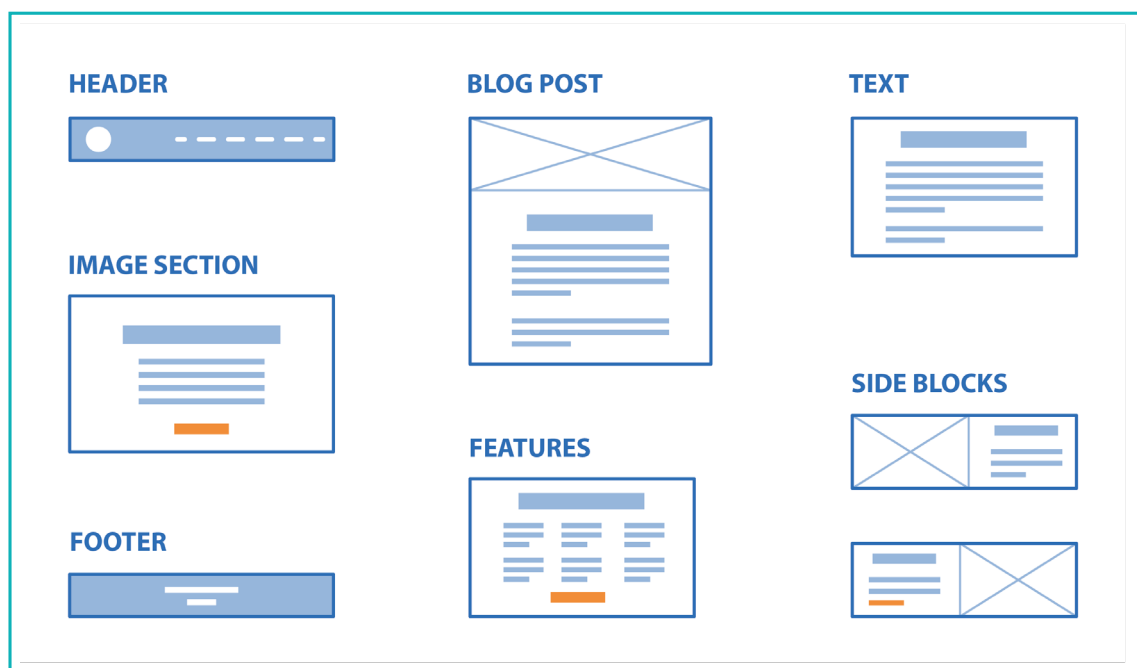


Image Reference - Droptica.com

Getting Started

NodeJS must be installed on our system before we can begin building our React app. Visit <https://nodejs.org/en/> and select the 'Download' button to download Node. You can install it by following the simple on-screen instructions once it has been downloaded.

After installing Node, we can begin developing our React application.

Note:

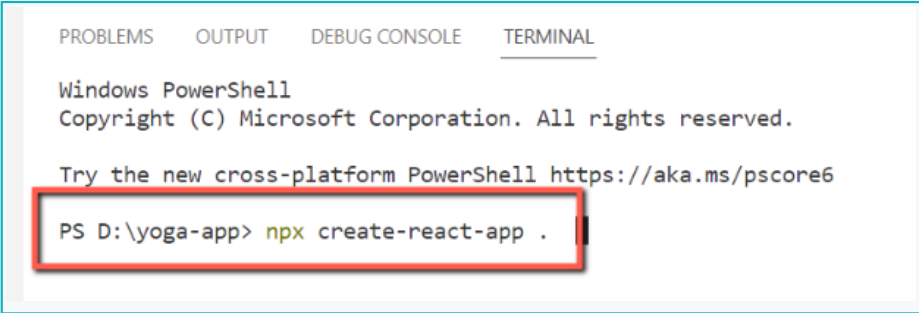
The primary objective of installing Node is to use the 'npm or Node Package Manager' for downloading the React library and create a React Project. Apart from that, Node serves no purpose in building our website using React.

Activity

Step 1: Creating a React Application

- 1 In VS Code, create a folder called "yoga-app". To start a React project, open a terminal in VS Code and type the command "npx create-react-app".

'npx create-react-app' is a command that downloads all the necessary files for a React app and then creates the React project for us. It is a simple but useful command that performs all the tasks associated with creating and configuring a React Project.



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS D:\yoga-app> npx create-react-app .
```

Note:

The command is terminated by a .(dot). The 'dot' indicates that the React project should be created in the same folder. If you want to use a different folder, simply replace the dot with the folder name.

Activity

- 2 The installation will take a few minutes, depending on your internet speed. When it is done, the terminal will display the following message. Your React app has been successfully installed.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

npm run build
  Bundles the app into static files for production.

npm test
  Starts the test runner.

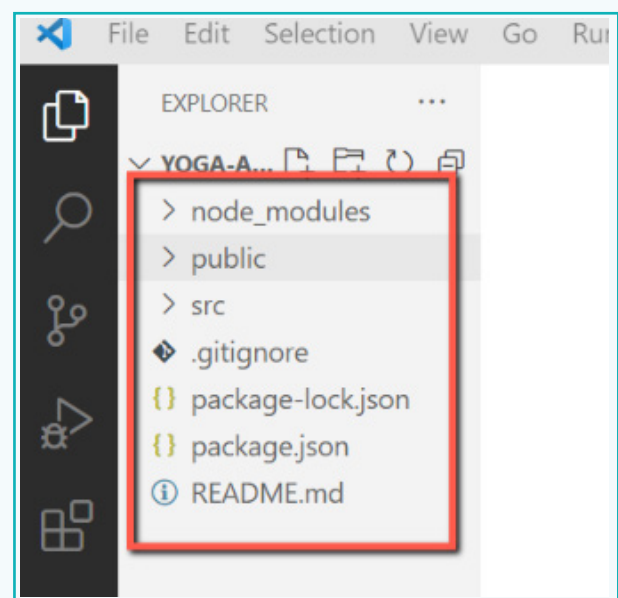
npm run eject
  Removes this tool and copies build dependencies, configuration files
  and scripts into the app directory. If you do this, you can't go back!

We suggest that you begin by typing:

  cd D:\yoga-app
  npm start

Happy hacking!
PS D:\yoga-app> 
```

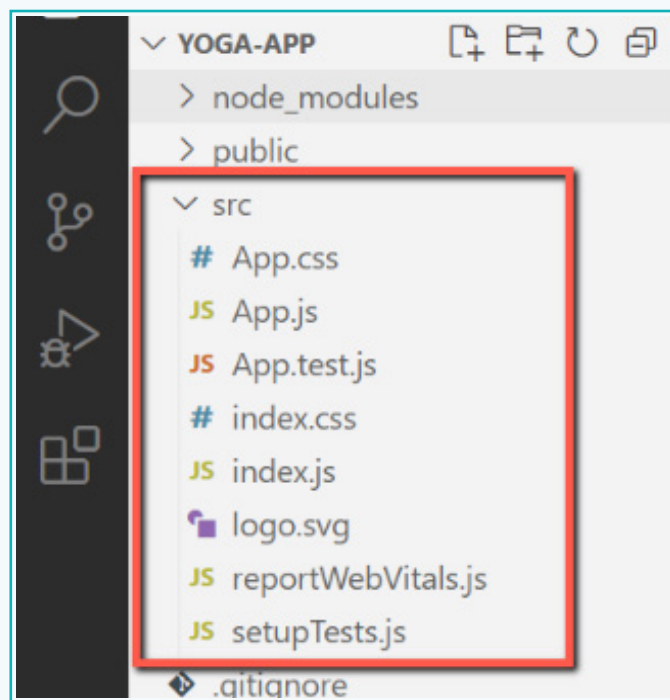
- 3 The project folder will contain several files and folders once the project has been created. The 'npx' command that we just ran will create the necessary files and folders we need for our React application.



Activity

- 4 The 'src' folder is the most important of all the files and folders, and we will be using the files within this directory throughout the project. A few notable files are mentioned below:

App.js	This is the main component file or the root component. All other components that we will create need to be included inside this component.
App.css	This contains CSS properties for the 'App.js' file.
index.js	This is the main JavaScript file that is responsible for kick-starting our application, loading components, injecting them into the Html file, and so on.
index.css	This contains CSS properties for the 'index.js' file.



Note:

The files, namely, 'App.test.js', 'logo.svg', 'reportWebVitals.js', and 'setupTests.js' are required for testing purposes and we will not be using them in this application.

Exercise

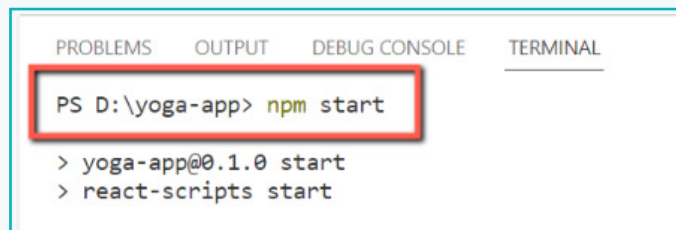
■ Fill in the blanks.

- 1 To create a react project, the command _____ is used.
- 2 In a React project, the root component is named as _____.

Activity

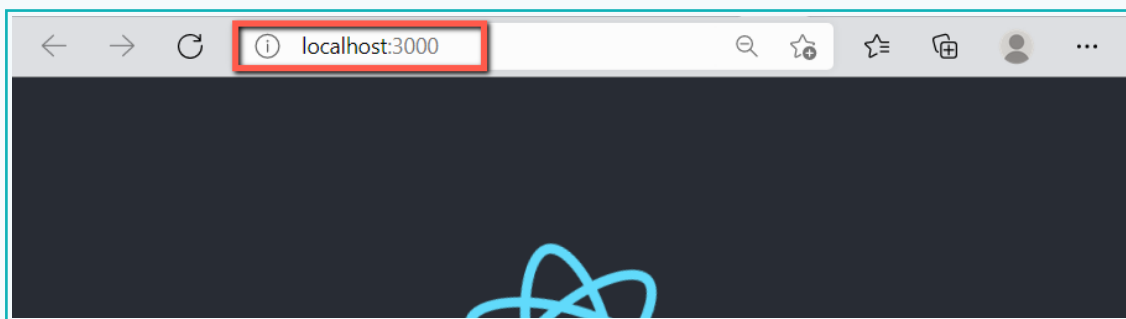
■ Step 2: Running/Executing the React application

- 1 Open your terminal and write the following command – “npm start”. This command will start a local server and execute the React application.



```
PS D:\yoga-app> npm start
> yoga-app@0.1.0 start
> react-scripts start
```

- 2 Output on the browser.



This is the default output that is displayed when we start any React application.

Exercise

State if the statements are True or False

3 The command 'npm run' is used to execute a React project.

▲ _____

4 React is a JavaScript library.

▲ _____

Activity

Step 3: Removing Unwanted Content from our Project

When you create a React project, some dummy content is automatically installed. Dummy content produces the default output that is displayed in the browser. Since we'll be starting from scratch, we will remove the default output and replace it with our own content. Getting rid of dummy content keeps your app clean and organised.

1 Inside the src directory, in the App.js file, remove the following code as indicated by the red border.

```
src > 16 App.js
1  import logo from './logo.svg';
2
3  import './App.css';
4
5  function App() {
6    return (
7      <div className="App">
8
9        <header className="App-header">
10         <img src={logo} className="App-logo" alt="logo" />
11         <p>
12           Edit <code>src/App.js</code> and save to reload.
13         </p>
14         <a
15           className="App-link"
16           href="https://reactjs.org"
17           target="_blank"
18           rel="noopener noreferrer"
19         >
20           Learn React
21         </a>
22       </header>
23     );
24   }
25 }
```

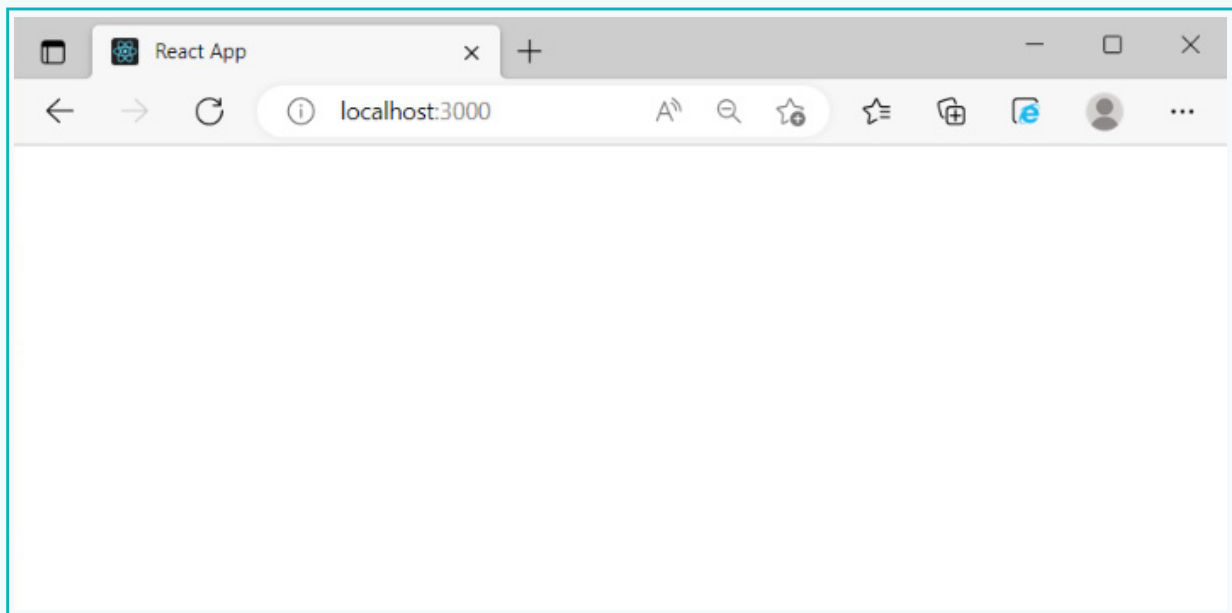

Activity

- 2 The 'App.js' file, after removing the default contents.

```
JS App.js
src > JS App.js > ...
1  import './App.css';
2
3  function App() {
4    return (
5      <div className="App">
6
7      </div>
8    );
9  }
10
11 export default App;
```

Note:

If you check the browser now, you will see an empty page because we have removed all the contents.

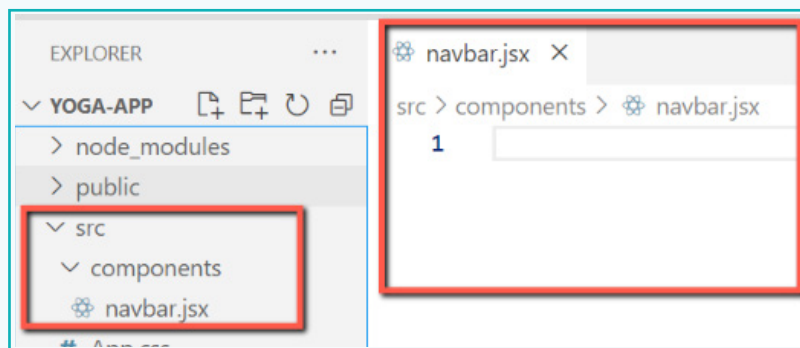


Activity

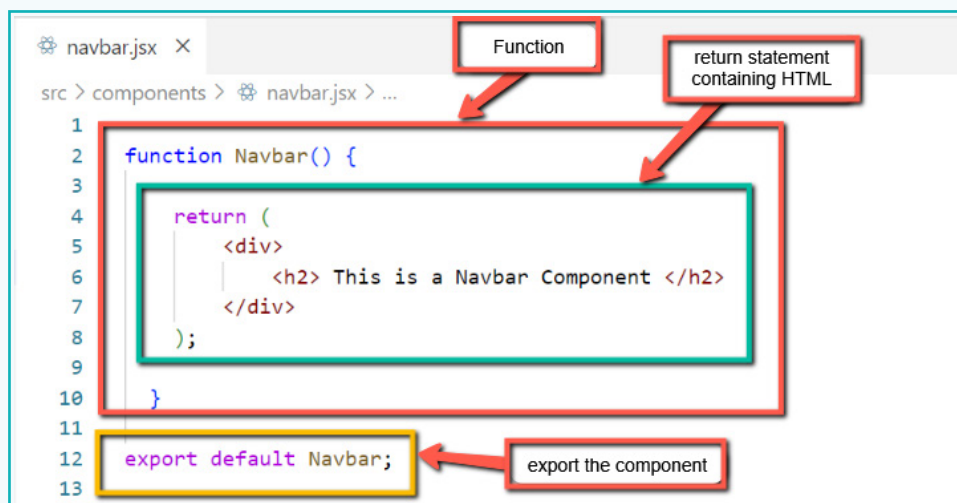
Step 4: Creating a Navigation Bar (Navbar) Component

Let us start building our application by creating a component.

- 1 First, we will make a 'navbar' component to represent the navigation bar. Create a folder called "components" in the 'src' folder to store our components. Inside the 'components' folder, create a file and name it "navbar.jsx".



- 2 In React, each component has its own unique structure.
- 3 First, we need a function that shares the same name as the component. A 'return' statement must be defined within the function. Inside this 'return' statement, we will use HTML to define the component's contents.
- 4 Secondly, we need an 'export' statement directive that allows us to access the component from other application-related files.



What is JSX?

If you look at the 'Navbar' component we just created, you will observe that we have inserted both HTML and JavaScript code in it. React provides JSX to enable this. **JSX** is an acronym for JavaScript and XML, which allows us to write HTML code directly within the JavaScript code. To use 'JSX' in a file, simply specify the file extension as '.jsx'. **Example** – navbar.jsx.

Exercise

■ Fill in the blanks.

- 5 'JSX' stands for _____ .
- 6 To make a component accessible to other components within the application, we use the _____ statement.

Activity

Step 5: Using the Navbar Component

The Navbar component has been created but has not yet been used anywhere. Whenever we create a component in React, we need to add it to our 'App.js' file in order to render it. The 'App.js' is the main or the root component of React and it acts as a container for all the other user-created components.

- 1 To use a component, we must first import it using the 'import' statement, and then, use it with a 'return' statement.

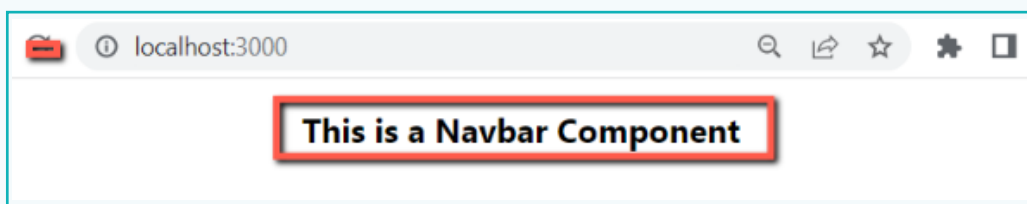
```
src > JS App.js > ...
1  import './App.css';
2
3  import Navbar from './components/navbar'
4
5  function App() {
6    return (
7      <div className="App">
8
9        <Navbar />
10
```

Activity

- 2 Now, let us check the output on the browser. Open the terminal and execute the command 'npm start'.

Note:

If your React application is already running, we do not need to type the 'npm start' command on the terminal.



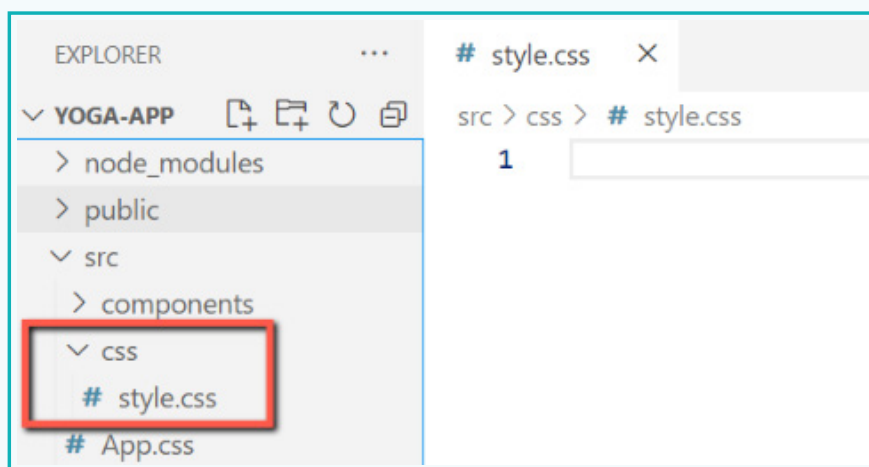
- 3 We can now remove the dummy content and add the following content, as shown in the image, now that we have verified that our application is functioning properly.

```
src > components > navbar.jsx > ...
1  function Navbar() {
2    return (
3
4      <nav>
5        <ul>
6          <li> HOME</li>
7          <li> YOGA POSES </li>
8        </ul>
9      </nav>
10
11    );
12  }
13
14  export default Navbar;
```

Activity

Step 6: Designing the component using CSS

- 1 To get started with CSS, we will create a folder named “css” inside the ‘src’ folder. Within the ‘css’ folder, create a file and name it “style.css”.



- 2 Add the following CSS code to design the Navbar.

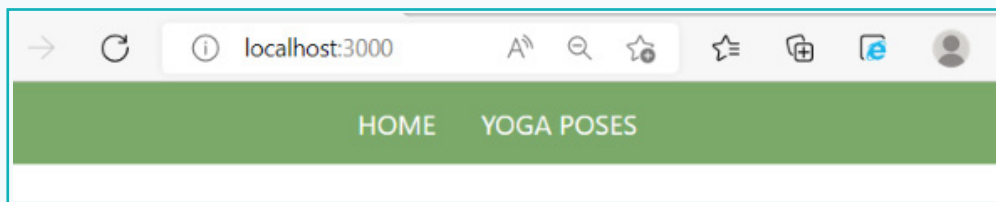


Activity

- 3 Import the 'style.css' file into the 'Navbar' component.

```
1
2 import '../css/style.css'
3
4 function Navbar() {
5   return (
6
7     <nav>
8       <ul>
```

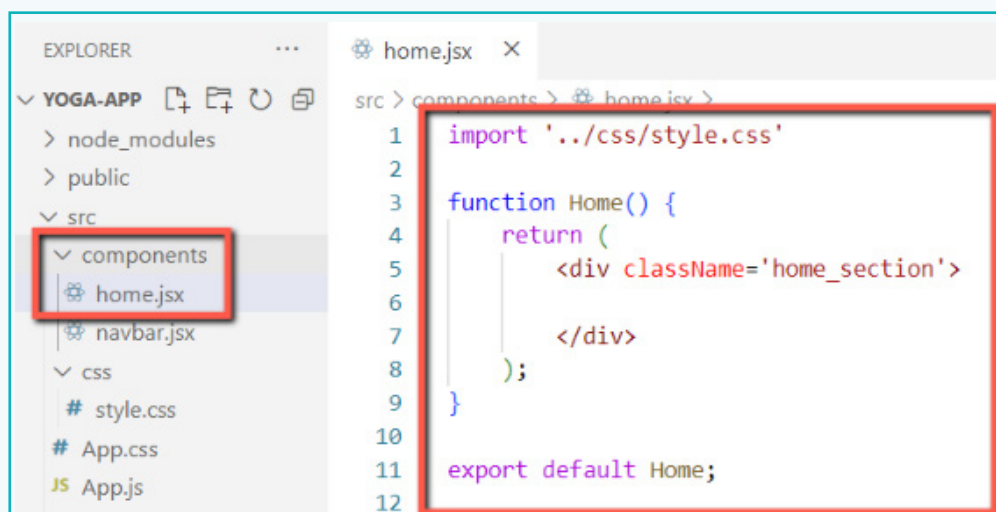
- 4 This is how the Navbar looks after applying the CSS.



Step 7: Creating the Home Component

Now that we are familiar with creating and using the components, we can proceed with adding the rest of the website.

- 1 Create another component and name it "home.jsx". Add the necessary 'React' component boilerplate code and within the 'return' statement, define a `<div>` tag and give it a class 'home_section'.



Activity

Note:

In React, to assign a class to an HTML element, the keyword 'className' is used instead of 'class'.

- 2 Inside the 'home_section' class, we will create another HTML container. Give it a class 'header' and add some content to it.

```
4
3  function Home() {
4    return (
5      <div className='home_section'>
6
7        <div className='header'>
8          <h1> YOGA FOR KIDS </h1>
9          <h3> The ancient science of adding Energy to Mind, Body and Soul.</h3>
10         </div>
11       </div>
12     );
13  }
```

- 3 Add the following CSS to this container:

```
src > css > # style.css > ...
20
21 .home_section {
22   width: 90%;
23   margin: auto;
24 }
25 .header {
26   height: 45vh;
27   background-image: url(../images/Untitled.png);
28   background-size: contain;
29   background-repeat: no-repeat;
30   text-align: left;
31   background-position: right;
32 }
33 .header h1 {
34   color: #7ba96a;
35   font-size: 3.2rem;
36   margin-top: 20px;
37 }
38 .header h3 {
39   color: grey;
40   font-size: 22px;
41 }
42
--
```

Activity

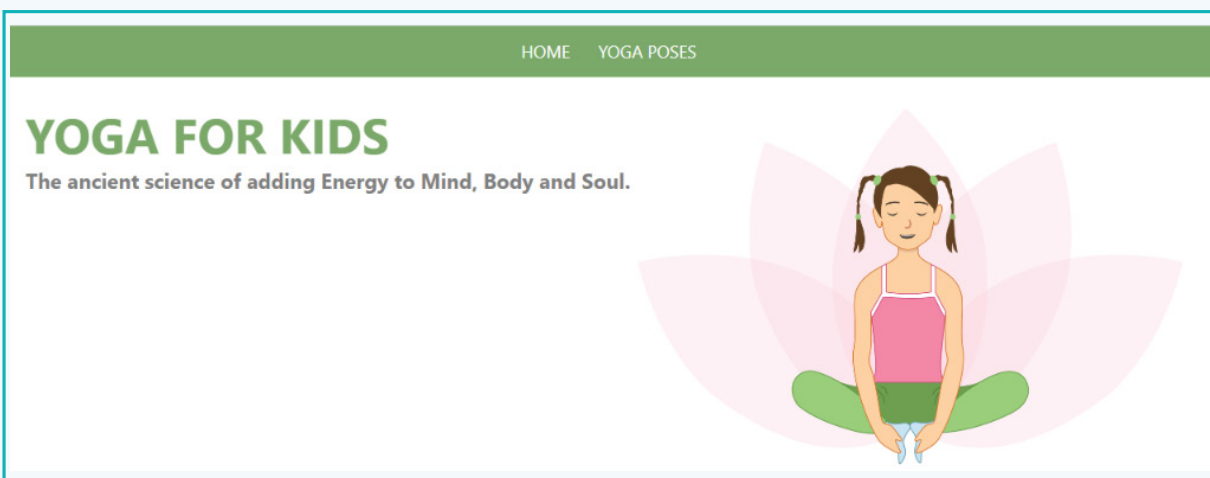
Note:

To add images, create a folder “image” in the ‘src’ folder and save the images in that folder.

- 4 Similar to how we added the ‘navbar’ component to the ‘App.js’ file, we will also add the ‘Home’ component to the ‘App.js’ file.

```
1  import './App.css';
2  import Navbar from './components/navbar'
3
4  import Home from './components/home'
5
6  function App() {
7    return (
8      <div className="App">
9        <Navbar />
10       <Home />
11     </div>
12   )
13 }
```

- 5 Output after adding the Home component.



Exercise

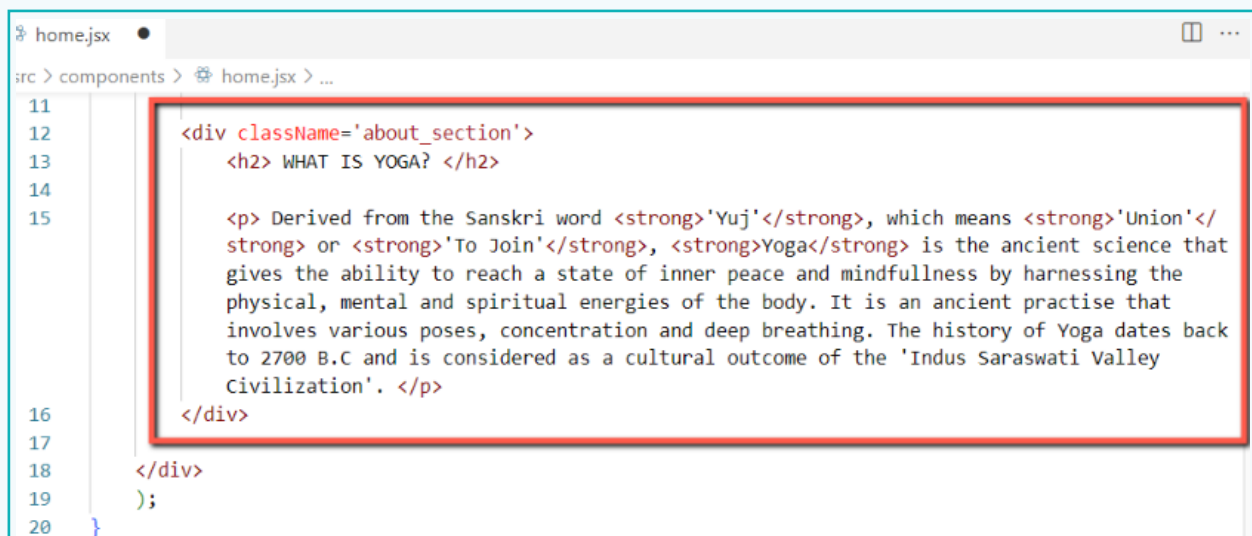
Fill in the blanks.

- 7 To add a class to an HTML element in a React application, the _____ property is used.
- 8 To add a component named “MyComponent” in the ‘App.js’ file., the correct syntax is _____ .

Activity

We will add two more `<div>` containers/sections in the ‘Home’ component.

- 6 Create another `<div>` container and give it a class ‘about_section’. In this section, we will add some details about the website.



```
home.jsx
src > components > home.jsx > ...
11
12 <div className='about_section'>
13   <h2> WHAT IS YOGA? </h2>
14
15   <p> Derived from the Sanskrit word <strong>'Yuj'</strong>, which means <strong>'Union'</strong> or <strong>'To Join'</strong>, <strong>Yoga</strong> is the ancient science that gives the ability to reach a state of inner peace and mindfulness by harnessing the physical, mental and spiritual energies of the body. It is an ancient practise that involves various poses, concentration and deep breathing. The history of Yoga dates back to 2700 B.C and is considered as a cultural outcome of the 'Indus Saraswati Valley Civilization'. </p>
16 </div>
17
18 </div>
19 );
20 }
```

Activity

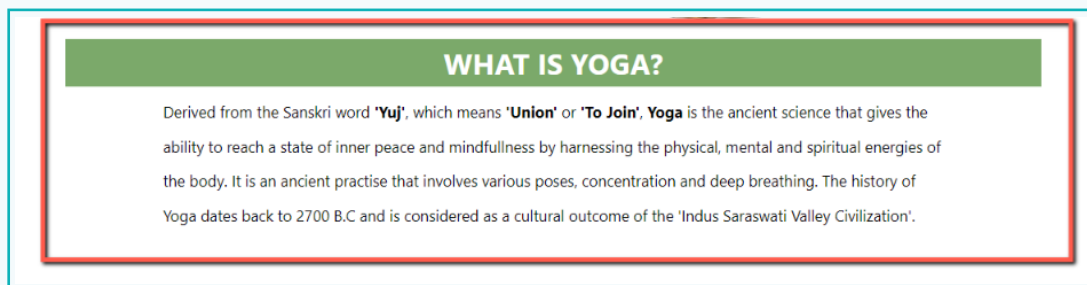
- 7 Add the following CSS to design this section.

```

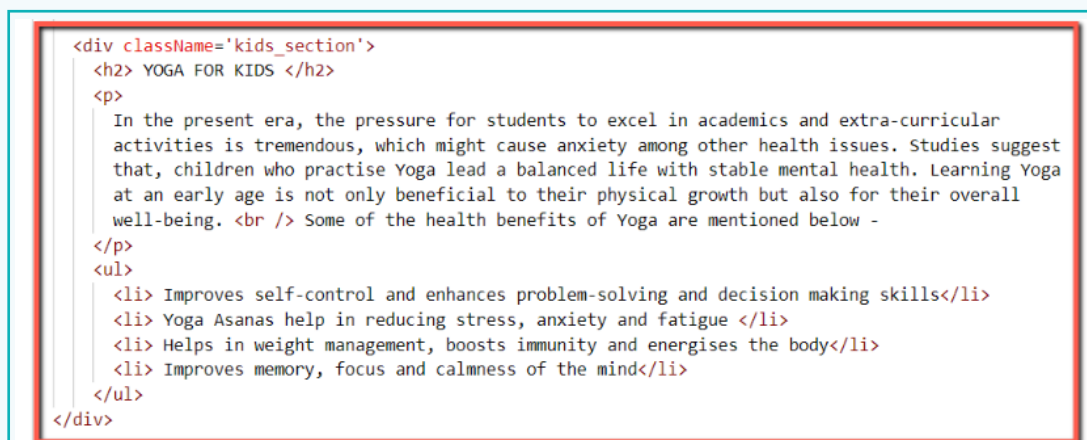
42
43 .about_section h2{
44     color: white;
45     font-size: 2.1rem;
46     background-color: #7ba96a;
47     padding: 5px;
48 }
49 .about_section p{
50     font-size: 18px;
51     width: 80%;
52     margin: auto;
53     margin-top: 10px;
54     line-height: 40px;
55     text-align: left;
56     margin-bottom: 20px;
57 }
58

```

- 8 Check the output before proceeding further.



- 9 Create another container/section using the `<div>` tag and give it a class 'kids_section'. In this container, we will provide information regarding the benefits of Yoga for kids.



Activity

- 10 The design that we will apply for this section will be the same as the 'about_section' section. So, instead of creating new CSS properties, we will simply add the class name 'kids_section' to the existing CSS design properties of the 'about_section' class.

```

42 .about_section h2, .kids_section h2{
43   color: white;
44   font-size: 2.1rem;
45   background-color: #7ba96a;
46   padding: 5px;
47 }
48
49 .about_section p, .kids_section p{
50   font-size: 18px;
51   width: 80%;
52   margin: auto;
53   margin-top: 10px;
54   line-height: 40px;
55   text-align: left;
56   margin-bottom: 20px;
57 }
58
        
```

YOGA FOR KIDS

In the present era, the pressure for students to excel in academics and extra-curricular activities is tremendous, which might cause anxiety among other health issues. Studies suggest that, children who practise Yoga lead a balanced life with stable mental health. Learning Yoga at an early age is not only beneficial to their physical growth but also for their overall well-being.

- 11 Let us add a few more CSS properties to design the list we have created.

```

59 .kids_section ul {
60   width: 80%;
61   margin: auto;
62   margin-top: 20px;
63   text-align: left;
64 }
65
66 .kids_section ul li {
67   font-size: 18px;
68   padding-bottom: 15px;
69 }
70
        
```

Learning Yoga at an early age is not only beneficial to their physical growth but also for their overall well-being.

Some of the health benefits of Yoga are mentioned below -

- Improves self-control and enhances problem-solving and decision making skills
- Yoga Asanas help in reducing stress, anxiety and fatigue
- Helps in weight management, boosts immunity and energises the body
- Improves memory, focus and calmness of the mind

This is how the 'Home' page should look if everything is done accordingly.

[HOME](#) [YOGA POSES](#)

YOGA FOR KIDS

The ancient science of adding Energy to Mind, Body and Soul.

WHAT IS YOGA?

Derived from the Sanskrit word 'Yuj', which means 'Union' or 'To Join', **Yoga** is the ancient science that gives the ability to reach a state of inner peace and mindfulness by harnessing the physical, mental and spiritual energies of the body. It is an ancient practise that involves various poses, concentration and deep breathing. The history of Yoga dates back to 2700 B.C and is considered as a cultural outcome of the 'Indus Saraswati Valley Civilization'.

YOGA FOR KIDS

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Some of the health benefits of Yoga are mentioned below -

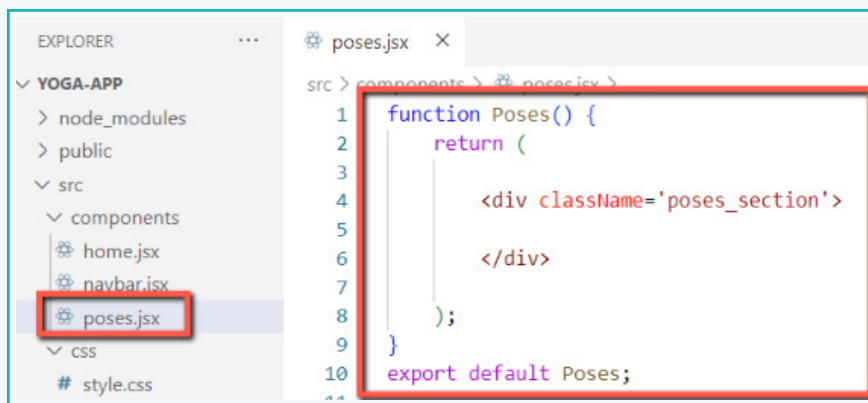
- Improves self-control and enhances problem-solving and decision making skills
- Yoga Asanas helps in reducing stress, anxiety and fatigue
- Helps in weight management, boosts immunity and energises the body
- Improves memory, focus and calmness of the mind

Activity

Step 8: Creating the 'Poses' Component

The purpose of creating this component is to display the various Yoga Poses, along with their brief description. The process for creating the component is the same, as described above.

- 1 In the 'Components' folder, create a file "poses.jsx". Add the basic component code to the file and inside the 'return' function, create a `<div>` container and give it a class 'poses_section'.



```
1 function Poses() {  
2   return (  
3  
4     <div className='poses_section'>  
5  
6     </div>  
7  
8   );  
9 }  
10 export default Poses;
```

- 2 Add a heading "Yoga Poses" inside the 'div' container and create another `<div>` container with class 'yoga_pose'.



```
1 function Poses() {  
2   return (  
3     <div className='poses_section'>  
4  
5       <h1> Yoga Poses</h1>  
6       <div className='yoga_pose'>  
7  
8       </div>  
9  
10    </div>  
11  )
```

Activity

- 3 In this container, we will add the details for a yoga pose. To put the details in an organized way, we will add the details inside another container. Also, add a class 'pose_details' to it.

```
<div className='yoga_pose'>
  <div className='pose_details'>
    <h3> Sanskrit Name - Sukhasana </h3>
    <h3> English Name - Easy Sitting Pose </h3>
    <p> <strong> Description </strong> - This is one the easiest poses to do. This pose
    involves sitting on the ground with your back straight, while keeping your head,
    neck and spine aligned. Now, cross your legs, bringing your knees towards the
    ground. The hands can be kept on the knees facing up or at together at the heart
    center </p>
  </div>
</div>
```

- 4 Now, we will also add an image here. However, to display images in 'React', we must first import them into the component and then use it as a value of the 'src' attribute.

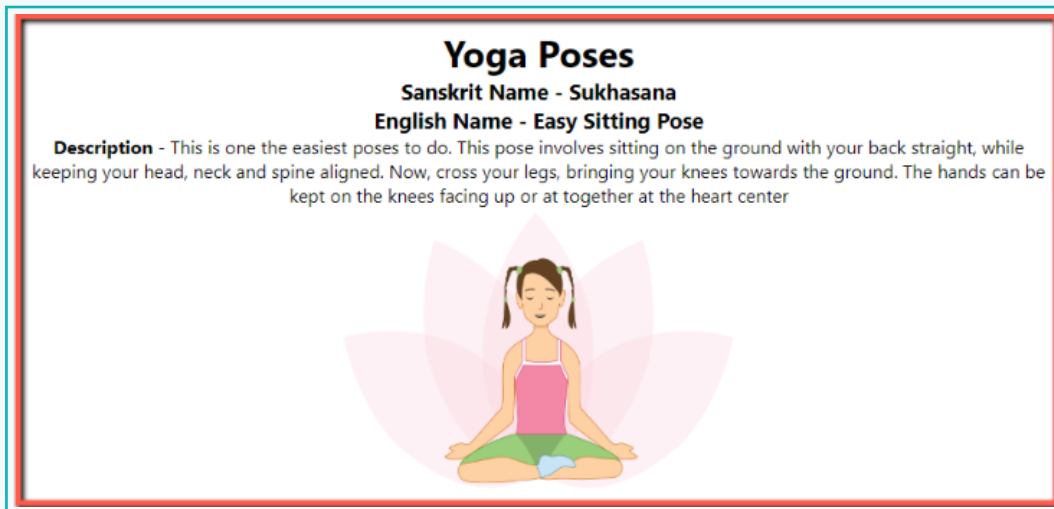
```
2
3 import pose_1 from '../images/Easy-Pose-1.png'
4
5 function Poses() {
6   return (
7     <div className='poses_section'>
8       <h1> Yoga Poses</h1>
9       <div className='yoga_pose'>
10        <div className='pose_details'>
11          <h3> Sanskrit Name - Sukhasana </h3>
12          <h3> English Name - Easy Sitting Pose </h3>
13          <p> <strong> Description </strong> - This is one the easiest poses to
do. This pose involves sitting on the ground with your back straight,
while keeping your head, neck and spine aligned. Now, cross your legs,
bringing your knees towards the ground. The hands can be kept on the
knees facing up or at together at the heart center </p>
14        </div>
15        <img src={pose_1} alt="" width='350px' height= '250px' />
16      </div>
17    </div>
18  )
19}
```

Imported image used here

Image Imported

Activity

- 5 The output looks like this:



- 6 Apply CSS to design the content of the component.

```
69
70 .poses_section h1{
71     color: white;
72     font-size: 2.1rem;
73     background-color: #7ba96a;
74     padding: 5px;
75     width: 90%;
76     margin: auto;
77 }
78 .yoga_pose {
79     width: 70%;
80     margin: auto;
81     display: flex;
82     justify-content: space-between;
83     box-shadow: 0px 3px 6px -1px rgba(0,0,0,0.75);
84     padding: 20px;
85     margin-top: 20px;
86 }
87 .pose_details {
88     text-align: left;
89     width: 60%;
90     line-height: 40px;
91 }
92
```

Activity

- 7 To make the CSS styles visible, import the 'style.css' file into the 'poses.jsx' file.

```
# style.css poses.jsx X
src > components > poses.jsx > Poses
1
2 import '../css/style.css'
3
4 import pose_1 from '../images/Easy-Pose-1.jpg'
5
6 function Poses() {
7   return (
```

- 8 Output after applying the CSS properties.

Yoga Poses

Sanskrit Name - Sukhasana

English Name - Easy Sitting Pose

Description - This is one the easiest poses to do. This pose involves sitting on the ground with your back straight, while keeping your head, neck and spine aligned. Now, cross your legs, bringing your knees towards the ground. The hands can be kept on the knees facing up or at together at the heart center



Exercise

■ Tick mark ☒ the correct answer

- 9 Which of the following CSS properties can have the 'flex' value?
- | | | | |
|---------------|--------------------------|-------------|--------------------------|
| a. Display | ☐ | b. Position | ☐ |
| c. Border-box | ☐ | d. Padding | ☐ |
- 10 Which of the following values belong to the 'justify-content' property?
- | | | | |
|-----------------|--------------------------|------------------|--------------------------|
| a. Block | ☐ | b. Space-between | ☐ |
| c. Space-center | ☐ | d. Middle | ☐ |

Activity

Step 9: Inserting More Yoga Poses

Let us add a few more Yoga Poses inside the component. We will rebuild the 'div' container with the class 'yoga_pose' for each yoga pose we create.

```
6 function Poses() {  
7   return (  
8     <div className='poses_section'>  
9       <h1> Yoga Poses</h1>  
10      <div className='yoga_pose'>  
11        <div className='pose_details'>  
12          <h3> Sanskrit Name - Sukhasana </h3>  
13          <h3> English Name - Easy Sitting Pose </h3>  
14          <p> <strong> Description </strong> - This is one the  
15        </div>  
16        <img src={pose_1} alt="" width='350px' height= '250px' />  
17      </div>  
18      <div className='yoga_pose'>  
19        <div className='pose_details'>  
20          <h3> Sanskrit Name - Sukhasana </h3>  
21          <h3> English Name - Easy Sitting Pose </h3>  
22          <p> <strong> Description </strong> - This is one the  
23        </div>  
24        <img src={pose_1} alt="" width='350px' height= '250px' />  
25      </div>  
26    </div>  
27  )  
28 }
```


Activity

- 1 In this container, create another 'div' with class 'pose_details' and add the details as we had done earlier. You will also need to import the image you want to display along with the yoga pose.

```
import pose_2 from '../images/Balasana.png'

function Poses() {
  return (
    <div className='poses_section'>
      <h1> Yoga Poses</h1>
      <div className='yoga_pose'>
        <div className='pose_details'>
          <h3> Sanskrit Name - Sukhasana </h3>
          <h3> English Name - Easy Sitting Pose </h3>
          <p> <strong> Description </strong> - This is one the easiest poses to do. This pose involves sitting on the ground with your back straight, while keeping your head, neck and spine aligned. Now, cross your legs, bringing your knees towards the ground. The hands can be kept on the knees facing up or at together at the heart center </p>
        </div>
        <img src={pose_1} alt="" width='350px' height= '250px' />
      </div>
      <div className='yoga_pose'>
        <div className='pose_details'>
          <h3> Sanskrit Name - Balasana </h3>
          <h3> English Name - Child's Pose </h3>
          <p> <strong> Description </strong> - This position mimics the natural body position of foetus in the womb. This pose involves kneeling down on the floor & sitting on your heels, keeping your big toes touching each other. Place your knees hip-width apart and inhale. Now bend forward and exhale as you lay your torso between your thighs and swing your hands forward. Hold this position for 2-3 minutes. To release, breathe-in slowly and return to the starting position. </p>
        </div>
        <img src={pose_2} alt="" width='350px' height= '250px' />
      </div>
    </div>
  )
}
```

- 2 Output after adding the second yoga pose.

Sanskrit Name - Sukhasana

English Name - Easy Sitting Pose

Description - This is one the easiest poses to do. This pose involves sitting on the ground with your back straight, while keeping your head, neck and spine aligned. Now, cross your legs, bringing your knees towards the ground. The hands can be kept on the knees facing up or at together at the heart center



Sanskrit Name - Balasana

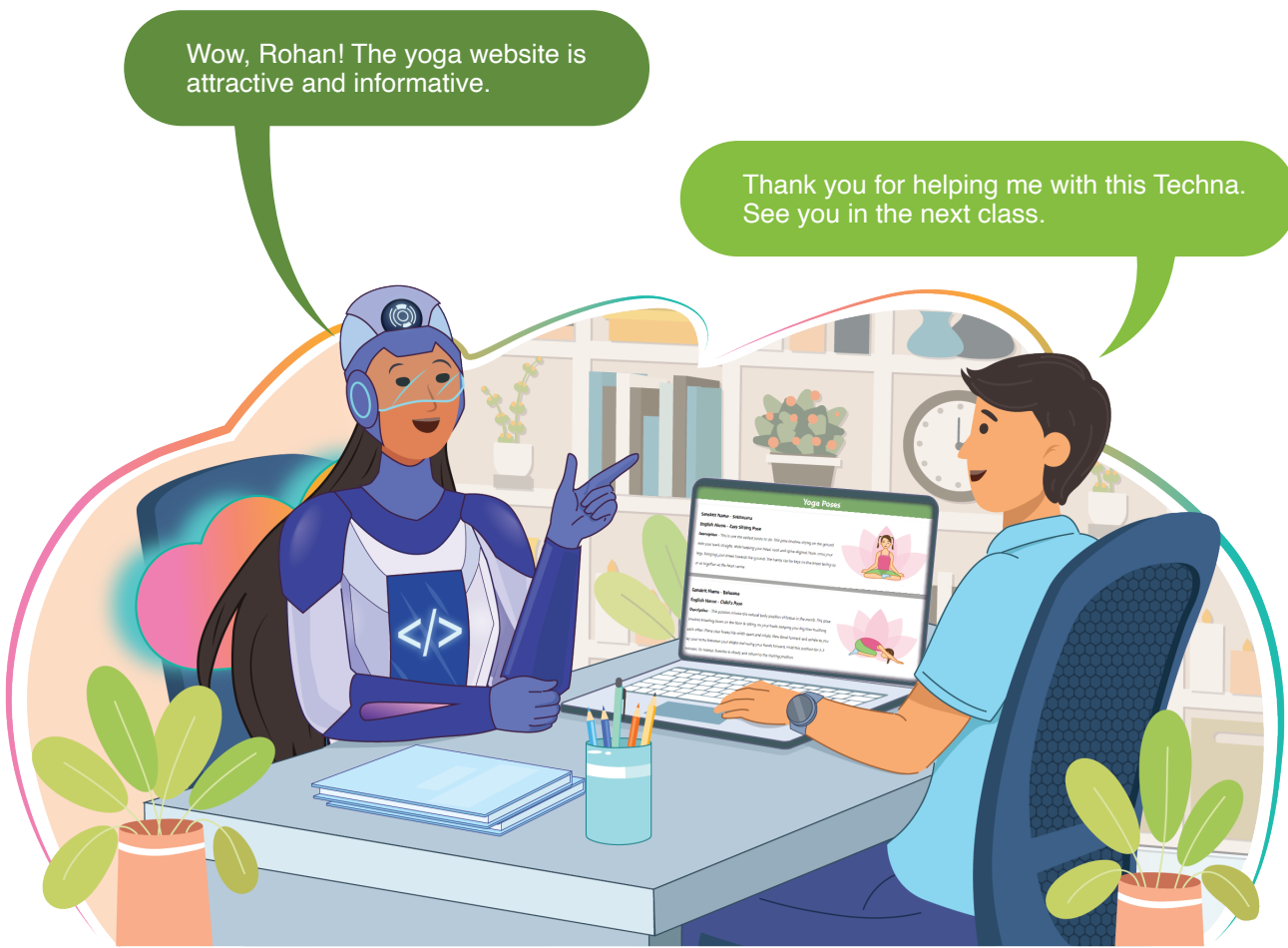
English Name - Child's Pose

Description - This position mimics the natural body position of foetus in the womb. This pose involves kneeling down on the floor & sitting on your heels, keeping your big toes touching each other. Place your knees hip-width apart and inhale. Now bend forward and exhale as you lay your torso between your thighs and swing your hands forward. Hold this position for 2-3 minutes. To release, breathe-in slowly and return to the starting position.



In this chapter, we looked at how to create user interfaces using the 'React' components. First, we learned how to create a 'React' application followed by its execution. We removed unnecessary content from our project before using the Navbar component to place the navigation bar. Then, we used CSS to design various components, such as 'home' and 'pose' components. Finally, we began to insert more poses.

In the same manner, you can add as many yoga poses as you like by creating the required 'div' containers and adding the necessary information to them.



Tech Flash

- **JavaScript Library** : A collection of pre-written blocks of JavaScript code blocks that facilitate the development of JavaScript-based applications.
- **REACT** : A JavaScript library, developed by Facebook to simplify the development of User Interfaces by breaking them down into several independent components.